

Zhiyi Chen

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Klaus Advanced Computing Building, Room 3337, Atlanta, GA, USA

Education

Georgia Institute of Technology

Atlanta, GA

Ph.D. Candidate in Computer Science; Advisor: Alberto Dainotti

Jan 2021 - Dec 2026 (expected)

Thesis: Leveraging ML and AI for a multi-dimensional understanding of Internet Autonomous Systems

Master of Science in Computer Science; Concentration: Machine Learning

Jan 2021 - May 2025

GPA: 4.0/4.0

Tsinghua University

Beijing, China

Bachelor of Science in Computer Science and Technology

Aug 2016 - Jul 2020

GPA: 3.73/4.0

Work Experience

ByteDance, Inc.

San Jose, CA

Research Intern; Advisor: Peng Li

May 2026 - Aug 2026

- Leading the development of a production-oriented Codex-based data-agent runtime for ByteDance's multimodal database, extending Codex with database-aware planning, skill routing, schema exploration/filtering, multimodal data exploration, and semantic-operator reasoning.
- Designed reliability and safety layers for sandboxed execution, SQL read/write policy, failure classification, bounded retry, audit tracing, and modular sub-agent orchestration for SQL critique and result verification.

Google LLC

Remote

Student Researcher; Advisors: Alex Krentsel, Sylvia Ratnasamy

Jan 2026 - May 2026

- Served as a main contributor to an internal multi-agent risk-assessment system for planned network changes, built with Google Agent Development Kit (ADK).
- Designed multi-agent workflows to jointly analyze heterogeneous internal documents, reports, and network telemetry for historical outage analysis and planned-change risk assessment.

ByteDance, Inc.

San Jose, CA

Research Intern; Advisors: Peng Li, Tieying Zhang

May 2025 - Aug 2025

- Developed TAHOE, a closed-loop execution-feedback learning framework for Text-to-SQL agents that implements an agent data flywheel by converting SQL execution traces into reusable experience through reflection, hint generation, retrieval, and attribution-based hint-bank management.
- Improved execution pass rate on the Spider2.0-Snow benchmark from 35.4% to 83.0% (+47.6 pp), reduced SQL debugging iterations by 11 $\times$, and demonstrated transferability across multiple LLM backbones.

Catchpoint Systems, Inc.

Remote

CTG Intern; Advisor: Gavin Lynch

Jun 2024 - Aug 2024

- Analyzed global Internet test data from measurement nodes targeting cloud and content providers.
- Designed automatic approaches for detecting false positives in inferred cloud outages and developed methods for detecting cloud performance degradations.

Coalition, Inc.

Remote

Research Intern; Advisors: Daniel Woods, Morgan Hervé-Mignucci

Jun 2023 - Aug 2023

- Leveraged public Internet measurement platforms and GIS tools to infer and visualize geographic locations of data centers that host policyholder domains.
- Explored the aggregation of cyber risks on physical infrastructure, especially critical points such as datacenters.

Publications & Preprints

- Chen, Z.,** Song, J., Li, P. "TAHOE: Text-to-SQL with Automated Hint Optimization from Experience." arXiv preprint arXiv:2606.12387, 2026. Under submission to ACM SIGMOD 2027. (LLM Agents for Databases)
- Chen, Z.,** Bischof, Z. S., Testart, C., Dainotti, A. "AS2Biz: Leveraging Web Presence and AI to Improve AS Business Classification", accepted at Internet Measurement Conference (IMC) 2026.
- Chen, Z.,** Bischof, Z. S., Testart, C., Dainotti, A. "Rethinking and Facilitating How We Classify Autonomous Systems by Network Properties." Invited for one-shot revision at the Internet Measurement Conference (IMC) 2026. (ML for Internet measurement)

- **Chen, Z.,** et al. “*AS2Org-Resolve: A Multi-Agent AI Pipeline for AS-to-Organization Mapping*”. Under review at ACM Internet Measurement Conference (IMC) 2026.
- **Chen, Z.,** Bischof, Z. S., Testart, C., Dainotti, A. “*Improving the Inference of Sibling Autonomous Systems*”, published in the Passive and Active Measurement Conference (PAM) 2023.
- Li, P., **Chen, Z.,** Chu, X., Rong, K. “*DiffPrep: Differentiable Data Preprocessing Pipeline Search for Learning over Tabular Data*”, published at ACM SIGMOD 2023. (ML for data cleaning)
- **Chen, Z.,** Lin, T. “*Principal Gradient Direction and Confidence Reservoir Sampling for Continual Learning*”, published in the International Conference on Artificial Neural Networks (ICANN) 2021.

Invited Talks

- “*Improving the Inference of Sibling Autonomous Systems*”, North American Network Operators' Group (NANOG) 87, Feb 2023, Atlanta.

Fellowships & Awards

- **Philip and Dianne Enslow Fellowship**, College of Computing, Georgia Institute of Technology, 2026
- **Best Community Artifact Award**, Passive and Active Measurement Conference (PAM), 2023, for the artifact accompanying “*Improving the Inference of Sibling Autonomous Systems*”.